

Investigating the Predatory Stress Response of *Hemigrapsus oregonensis* to the *Carcinus maenas* in Warming Environments – Chase Moody

ABSTRACT

Across Washington coasts, the highly invasive European Green Crab (*Carcinus maenas*) has taken root and caused ecosystem degradation. Intertidal regions and coastal waterways here are also at immense risk from future increases in SST. The Hairy Shore Crab (*Hemigrapsus oregonensis*) is a common intertidal species frequently preyed upon by the European Green Crab. While both are strong thermoregulators, there is little knowledge into the combined response of temperature and predatory stress. This study examines whether increased temperature affects behavioral and physiological stress responses in the HSC in response to the EGC. Here we show that there is minimal difference across temperature treatments in predatory response. This study highlights little evidence high or low temperatures responding differently to predation and instead indicated a stronger temperature stress and metabolic depression effect. This is concurrent with physiological understandings but contradicts initial predictions of more stress in higher temperature individuals. This indicates that overall health, thermal stress, and environmental conditions in shore crabs are better indicators of physiological strength and fitness than a brief predatory stimulus. Shore Crabs do not have enhanced physiological sensitivity towards predation in response to overall environmental health factors. This study serves as a continuation and exploration of physiological limits in shore crabs and serves as a model for how environmental conditions have a strong selection for survivability and fitness. Further exploration into the conditions regarding metabolic depression and continued predator exposure, replicating this studies' continuous temperature and metabolic pressure, could provide more insight into interspecific dynamics and future ecosystem health and environments shift.

INTRODUCTION

From the Pacific to the Atlantic, from tip to tip of coastline, the destruction of the invasive European Green Crab (*Carcinus maenas*) runs rampant. While coastal communities are no stranger to invasive species, this particular decapod has had very noticeable and highly negative impacts on a variety of environments, including the coastal and intertidal regions of Washington State (Turner, 2025). Research has shown that these European Green Crabs have primarily preyed on the Hairy Shore Crab (*Hemigrapsus oregonensis*), a prevalent and vital link in lower trophic levels (Fisher, et al 2024). Past studies have found this shore crab to be well adapted to high temperature conditions (Todd, Dehnel, 1960). Similarly, research regarding Green Crab preferred habitats found a strong preference and correlation to higher temperatures (Nielsen, et al 2025), unfortunately indicating a strong

potential for increased invasive activity in warmer waters. With the Puget Sound, like many other waterways, likely experiencing noticeable temperature increases in the coming years (Walker, et al 2022), this invasive potential and environmental stress may get a better chance to manifest.

While most research has focused on physiological limits with respect to temperature and salinity, the distribution of crab species, and the ecological impacts of the invasive European Green Crab, little research has investigated the paired physiological response between predator stimulation and temperature increase in the Hairy Shore Crab. Given this gap in the primary literature and the increased outlook for both predator encounter and stressful environmental conditions, this study aims to investigate whether heat stress influences behavioral and physiological responses when exposed to the predatory European Green Crab. To accomplish this, we measured behavioral responses before

and after predator encounter, in multiple temperature conditions. Similarly, we measured respiration to monitor physiological stress across the same host of treatment groups.

METHODS

SAMPLE COLLECTION

Hairy Shore Crabs were obtained from coastal collection sites on the Kitsap Peninsula, near Bremerton. They were then fasted for the duration of the study length.

BASLINE MEASUREMENTS + SETUP

To begin the study, our Hairy Shore Crab (HSC) individuals first needed to be physically and behaviorally measured, to achieve a strong baseline value to compare with the coming high stress weeks. Weight, length, and other physical observations were recorded, including whether they were already missing limbs at the beginning of the study. These baseline measurements were used to standardize respiration by body mass. Various experimental procedures, listed below, were performed this first week as well. From here, the subdivided groups (3 males, 3 females for each temperature treatment) were placed in their respective holding tanks and placed in the proper water treatment. The elevated water temperature treatment was 24 Degrees Celsius, and the ambient water temperature treatment was 14 degrees Celsius. Salinity was held constant at 35ppt.

RESAZURIN ASSAY

The resazurin assay was used to measure baseline and induced stress response respiration. To accomplish this, we utilized the resazurin methods outlined by Ariana Huffmyer, in 'Blue Notes and Bivalves', 2025. We retained components like the resazurin makeup and blank procedure, but instead used a 6 well plate, with only 8ml total working solution in each well plate at their adjusted temperature (in this case 14C) (See picture 1). We also alternatively used 200uL aliquots for each time interval, placed into the 96 well plate. The time intervals were in 15-minute increments (0, 15, 30, 45). We limited our assay time to 45 minutes to retain consistency across weeks. RFU was computed

for each individual crab/well, and the raw level was adjusted per their body weight.

RIGHTING TIME + BEHAVIORAL MOVEMENT/PREDATOR ENCOUNTER

From here, in a standard fashion for all HSC, we measured righting time by placing them on their carapace and seeing how long it took for each crab to right itself. Clean surface area and consistent timing metrics were maintained. To quantify their baseline movement patterns and behaviors, we placed them into the large experimental tank (See Picture 2) with measured walls and distances and recorded their total movement over 30 seconds. All of the baseline behavioral and physiological were critical to record before our specimens had any temperature or predator stress. In Predator encounter weeks, the HSC were placed in the exact same location in the experimental tank, with the EGC now being on the other side of the mesh. Again, total horizontal movement was measured over 30 seconds, and the crabs were removed.

WEEKLY TREATMENTS

Week 6 conducted the baseline measurements, baseline respiration in 4 crabs from each treatment, righting time, and movement without predator encounter.

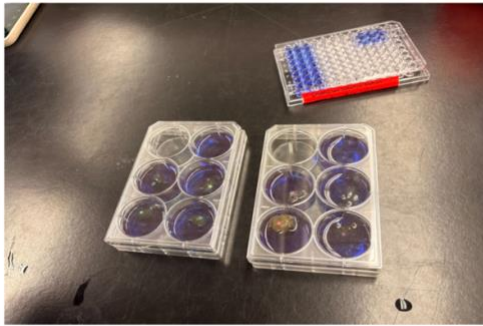
Week 7 conducted a preliminary respiration measurement in the same 4 crabs each treatment from week 6, then pre-predator righting time, as well as pre predator movement in the other 2 crabs for each sample. This was followed by movement with predator present in all samples, then post-predator righting time in all samples

Week 8 conducted initial righting time, and samples were exposed to predator. Horizontal distance could not be calculated here, due to experimental limitations and mortality. We then did a post-predator resazurin assay in the same crabs used for resazurin each week.

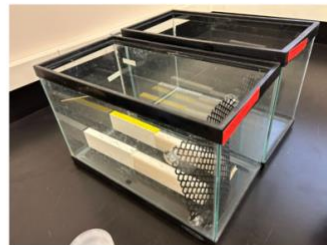
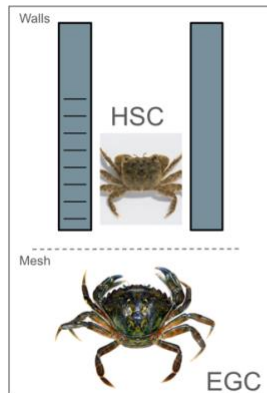
ANALYTICAL METHODS

Results were analyzed in RStudio. ANOVA, Tukey, and Mixed-Effects models were used for brief statistical interpretation and takeaways. ANOVA + Tukey were used to compare resazurin assays across weeks, and treatments, and the Mixed-Effects model was

used to take advantage of the repeated righting time measurements on the same crabs across different treatments and predator encounters.



Picture 1: Actualized Resazurin Setup. Seen here is the 6 well plates, blanks, active samples being recorded, and aliquots stored in 96 well plate above.



Picture 2: Idealized and Actualized Behavioral/Distance Traveled/Predator Encounter Tank. Walls were introduced to control horizontal movement, while mesh was used to allow olfactory sense detection while still protecting Hairy Shore Crab. Tick marks represent measurements (cm)

RESULTS

The results from these experimental setups were designed to answer the question of whether temperature treatment would influence stress response to a predator stimulus in the Hairy shore Crab, using the European Green Crab as the predator stimulus. In comparing temperature treatments before and after predator stimulus, as well as comparing it to baseline measurements, we can draw interpretations regarding their stress responses and differences.

Figure 1 depicts the Total Horizontal Movement in our samples across a few treatments. Unfortunately, we could not obtain any data for week 8, in the interest of the livelihood of our crabs. Week 6, before any temperature treatment or predator encounter, despite the small sample size, we saw very similar results across the crabs that would eventually be in their respective groups. We saw two types of profiles: a fast moving, intermittently stopping profile, and a slower moving profile with minimal total movement.

Understanding a consistent baseline was critical for the understanding going forward. Figure 1 also illustrates week 7, before and after predator encounter. Before the predator encounter, and after a week of temperature exposure, there were consistent profiles in the two temperature treatments. The elevated temperature treatment crabs saw rapid movement and were quicker to move than in week 6. We did not see this trend manifest in the ambient temperature, where their respective profiles from 6 remained relatively constant. In week 7, after the predator encounter, we saw relatively consistent profiles between the two treatments, with the same pattern of fast movers and slow movers, with similar distributions across treatments with respect to what profile they had. While there isn't a major noticeable difference between groups overall variability in the movement increased after predation stimulus.

Figure 2 illustrates the righting time in our samples across weeks and treatments. At the same time, it divides up week 7 into the pre-predator and post-predator righting times. In all sections except Week 7 Pre-Predator, there was no significant difference between temperature treatment groups. Typically, average values for righting time ranged from 0.75-1 seconds, consistently across the board. In week 7, before predator exposure, there was interestingly a significant difference between the two temperature groups. This was the only deviation we saw. It is important to note the metabolic stress over the course of weeks 7 and 8, and the immense handling stress associated with the week 6 setup.

Figure 3 considered respiratory levels across weeks, and after predator exposure in week 8. Week 6 was significantly different from the subsequent weeks, and respiration levels were much higher across the board. There were no differences across temperature groups within weeks. There was also no difference between weeks 7 + 8. It is important to note slight changes in resazurin setup between weeks 7 + 8, and potential solution issues per our blanks in week 8. Our elevated temperature group in week 8 was experiencing increased mortality. They also were incredibly active, trying to escape their wells.

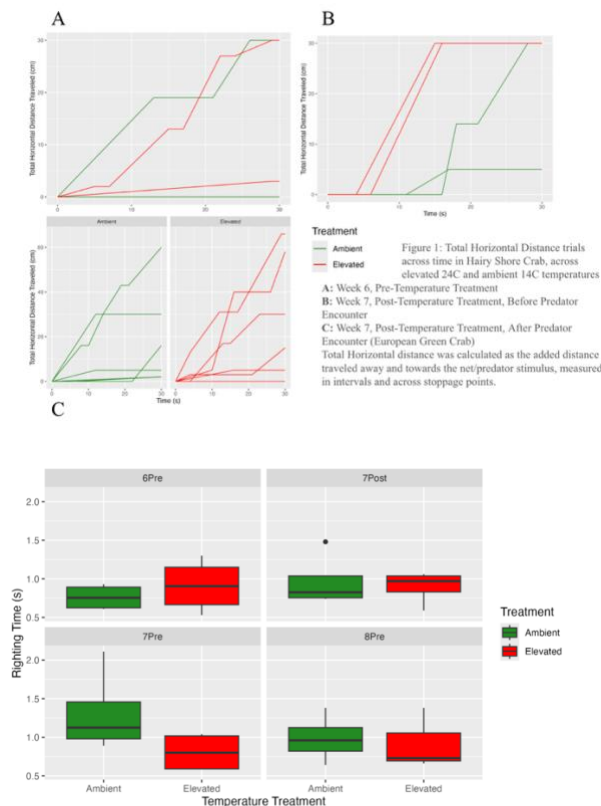


Figure 2: Boxplot illustrating righting time in Hairy Shore Crabs across weeks and treatments (elevated 24C and ambient 14C temperatures);

6Pre: Week 6, no predator exposure
 7Pre: Week 7, no predator exposure
 7Post: Week 7, after predator exposure (European Green Crab)
 8Pre: Week 8, no predator exposure

Brief Statistical Interpretation using Mixed-Effects Model yielded significant results ($p = 0.02$) in Week7Pre

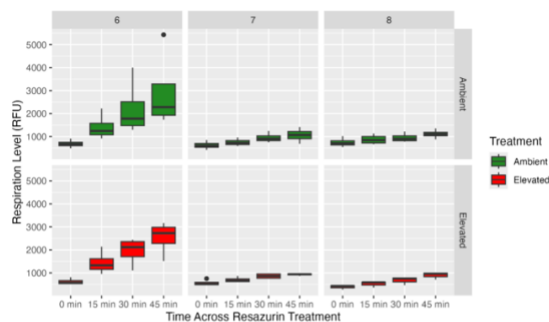


Figure 3: Boxplot reflecting average Resazurin RFU levels and Trials Across Weeks and Treatments (elevated 24C and ambient 14C temperatures) in Hairy Shore Crab.

Week 6 + 7: Pre-Predator Exposure
 Week 8: Post-Predator Exposure (European Green Crab)

Brief Statistical Interpretation using ANOVA and Tukey Tests yielded significantly different results comparing week 6 to week 7 ($p = 0.0000019$) and comparing week 6 to week 8 ($p = 0.0000021$)

DISCUSSION:

To begin, the data collection, quality of collection, and experimental errors should be considered. As mentioned before, the most critical change in our methods revolved around our rapidly perishing Hairy Shore Crabs. Tragically, at the beginning of week 8, two of our elevated temperature crabs had passed, and one passed away in the process of our sample collection. Because of this, and in the interest of getting usable resazurin data, we had to make the decision not to do post-predator righting time, and only briefly expose them to the predator, choosing not to record data there either. While this limited righting time and movement data, it gave us the ability to compare our pre-predator exposure resazurin data to post-predator resazurin data, which was critical for understanding the differences in treatments with respect to the predator. During this resazurin trial in week 8, we had the crabs attempt to crawl into other wells frequently, resulting in us having to correct their actions and potentially weakening data. These mortalities could have been caused by resazurin-induced physiological processes, as the compound is known to interfere with molting. Another point of confusion and experimental error we ran into was issue with the resazurin solution in week 8. While there was no statistically significant difference between the two temperature treatment groups, some of the wells with our crabs for our elevated temperature treatment had incredibly low RFU values. These values were below that of the blank, indicating potential contamination like mentioned above or some other unexplained biological process. It is important to note our incredibly low sample size as well. 2-6 samples might not be perfectly indicative of a population metric.

The main objective of this study was to determine whether temperature treatment would have influence on predatory stress response in Hairy Shore Crabs when exposed to European Green Crabs. Initially, we predicted that under extended heat exposure, crabs would exhibit higher levels of predatory-induced behavioral change in comparison to individuals

experiencing less thermal exposure. If this finding were to be supported, it might have implications for warming coastal ecosystems who are also highly susceptible to European Green Crab invasion. Specifically, if high-heat stress could exhibit noticeably more acute predatory stimulus when encountering predators in comparison to low-heat stress, we could infer that their success, survival, and fitness would decrease even more with the predator invasion. These additive and combined effects, with increased invasion, increased SST, and increased combined physiological responses could point towards the Hairy Shore Crab becoming even more predated on, and population numbers potentially being depleted.

In terms of our total movement data (Figure 1), we did not see any difference between temperature groups when combined with predatory behavior. While we did see an increase in erratic behavior, there was no difference between the two temperature groups. Very interestingly, while these profiles were very similar post-predator, the profiles for the HSC before predator stimulus were very different. Seemingly, the elevated temperature crabs were strongly correlated to quicker movement, showing a marked shift from their baseline. The ambient temperature group didn't really change from their baseline. This indicates that the initial temperature had a more profound effect on their movement dynamics between weeks 6 + 7 in comparison with predator stimulus for the elevated group.

We saw a similar trend in the righting time data (Figure 2). From week 6, where we saw similar profiles, to week 7 before the predator stimulus, there was a clear differentiation between temperature groups. The ambient group had a longer righting time, potentially indicating that they were less stressed for survival. We saw the same 'purifying' trend post-predator encounter in week 7. Both temperature groups had similar righting times after being exposed to the EGC. Again, this indicates that the initial temperature stress had different effects on the ambient temperature and elevated temperature groups, and that predator exposure had similar end results in righting time. This might indicate that the elevated group was already well above their baseline stress level

before predation as seen in their minimal change before and after EGC stimulus. This might indicate that, like with the movement, the ambient temperature group wasn't as stressed initially. Essentially, while both groups performed similarly with the predator, the ambient temperature group had the 'stress budget' to show an increased stress response. This might indicate the same initial temperature stress response being more dominant in the elevated crabs.

The respiration data (Figure 3) shows a similar trend in a slightly different way. Essentially, the main takeaway here was that there was no difference between pre-predator and post-predator respiration levels across weeks 7 + 8, but that the initial temperature treatments + physiological stress associated with the experimental conditions had an immense effect on respiration in both groups. This supports the temperature stress response with our elevated group, but because we see the same trend in the ambient group, it might be more indicative of an overall physiological stress. As there was no difference between temperature groups, we might assume that didn't have as large of an impact as the potential metabolic depression exhibited across the board.

Essentially, we saw a strong metabolic depression due to experimental setup and design across all treatment groups, and a noticeable stress increase in the increased temperature Hairy Shore Crabs that they maintained throughout predator interaction. The HSC in the ambient group saw the same high stress after predator interaction and in week 8 but showed a smaller spike in stress levels due to their temperature treatment earlier on. There was no clear difference in performance when considering just the physical and metabolic changes with EGC stimulus.

In comparing this to previous research in the field of shore crabs and increased stress response, there are some clear similarities and considerations. The most important consideration, purely experimentally, might come down to the fact that a shore crab likely wouldn't have to endure such high heat stress constantly. As strong thermoregulators, to be under immense heat stress for weeks would be very rare. One thing we did see was a sustained

level of manageable stress levels across predator and environmental conditions in our elevated temperature group, concurrent with research indicating strong environmental and thermoregulatory adaptability (Todd, Dehnel, 1960). In our ambient HSC, we did also see the very clear response in variable behavior to the EGC, potentially indicating an evolutionary or biological response to their well described prominent predator (Fisher, et al 2024). While we weren't able to pull apart differences in predatory response across temperature treatments, showing this stress maximum in both groups helps us inform our understanding of the physiological limits of this species, and their potential for metabolic and behavioral adaptability.

We also saw similarities to other shore crab research in terms of movement kinematics (Coles, et al 2024), with our ambient (colder) temperature treatment taking longer with their movement in general, especially seen in week 7 before predator.

While we didn't see differences directly within predatory stress responses across temperatures, we did see a marked effect with potential metabolic suppression across all treatments and a high level of continuous, maximum stress in our elevated temperature Hairy Shore Crabs. Future research could lean more into this fact and focus on the changes in overall metabolic activity over longer term exposure to predators. Clearly, the extended treatment had a profound effect in all crabs. In a different way, it would be interesting to see how this potential metabolic depression manifests in these types of conditions, potentially experimenting with food resources or increased distance from other individuals. These overall reductions may be reflecting stress-associated metabolic suppression, but the exact mechanics by which it manifests needs further exploration. It appears other research regarding metabolic depression focuses more on bacterial/viral infection, or with injury. Taking temperature out of the experimental process entirely or focusing more on intermittent temperature exposure more concurrent with intertidal regions could yield more ecologically significant results.

As temperatures rise and invasion increases, the combined effects of temperature

and predatory stress are sure to become more prevalent, while also experiencing their own evolutions and changes over time.

Understanding the susceptibility and physiological limits of vital communities like the Hairy Shore Crab could yield massive understanding for changing ecosystem dynamics and help preserve our intertidal regions. The understanding that temperature shock and overall metabolic depression (starvation) have more of an effect than predatory response implicates the importance of overall ecosystem health, while still considering invasive species.

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